

```

import java.util.*;

public class CharacterMapReduce {

    // Mapper
    public static List<String[]> mapper(String data) {
        List<String[]> mappedData = new ArrayList<>();
        for (char ch : data.toCharArray()) {
            if (Character.isLetter(ch)) {
                char lowerChar = Character.toLowerCase(ch);
                mappedData.add(new String[]{String.valueOf(lowerChar), "1"});
            }
        }
        return mappedData;
    }

    // Reducer
    public static Map<String, Integer> reducer(List<String[]> mappedData) {
        Map<String, Integer> counts = new TreeMap<>();
        for (String[] pair : mappedData) {
            String key = pair[0];
            int value = Integer.parseInt(pair[1]);
            counts.put(key, counts.getDefault(key, 0) + value);
        }
        return counts;
    }

    public static void main(String[] args) {
        // Given dataset
        String data = "Hello World Map Reduce is Easy Hello Python";

        // Map phase
        List<String[]> mappedData = mapper(data);

        // Reduce phase
        Map<String, Integer> result = reducer(mappedData);
    }
}

```

```
// Display output
System.out.println("Character Occurrences:");
for (Map.Entry<String, Integer> entry : result.entrySet()) {
    System.out.println(entry.getKey() + " : " + entry.getValue());
}
}
```

Output :

Character Occurrences:

a : 2

d : 2

e : 6

h : 2

i : 1

l : 4

m : 1

n : 2

o : 4

p : 2

r : 2

s : 1

t : 1

y : 2